Ricci-Flow Extension in Four Dimensions: Exact Collar Operators, Quadratic Compatibility, and Controlled Inexact Reconstruction

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Abstract

The September 24, 2026 manuscript on bounded scalar curvature and four-dimensional Ricci-flow extension asserts smooth continuation on the original closed manifold. Its collar construction is an analytic selection from an overdetermined Ricci operator, not an inverse of an arbitrary matrix and not a solution of arbitrary Ricci-flat boundary data. We examine all mathematical source files and isolate the exact range identity used by that construction. For the original weighted exterior spaces we prove a reflexive reduction, compute the actual quadratic Ricci term, and derive a third-order-accurate local collar predictor together with its second-order compatibility obstruction. We then prove residual-to-branch estimates, inner-boundary-preserving reconstruction bounds, and parameter derivative and exterior variation budgets. These are conditional on the source's bounded collar operator; they do not supply its global geometric or static inputs. The outer-boundary term, zero-mass cancellation, nontrivial quotient group, and order of geometric limits are retained. Actual local proofs, finite algebra checks, source claims, and unverified global dependencies are distinguished throughout. Contribution assessment: C2/5, source-specific local operator and nonlinear residual refinements with complete restricted proofs, without independent certification of the global extension theorem.

1 The theorem under examination and the logical scope

The source [1] considers a smooth Ricci flow

$$\partial_t g = -2\,\mathrm{Ric}(g), \qquad 0 \le t < T < \infty,$$

on a closed connected manifold of real dimension four. Its main assertion is that a uniform bound on the scalar curvature implies smooth extension past T on the *same* manifold. This is stronger than continuation from an orbifold terminal space. It also cannot be inferred just from small Ricci curvature on rescaled regions: a nonflat Ricci-flat bubble has zero Ricci tensor and nonzero full curvature.

This report does not replace that assertion by a weaker theorem and then call the original assertion verified. The new proofs below concern a fixed quotient exterior, a fixed weighted Banach space, and an explicitly specified collar selection operator. The source's concentration analysis, parabolic estimates, and imported inequality on degenerating Ricci-flat trees are indispensable

additional inputs for its global conclusion. Reading all their statements and proofs is not formal verification of the entire chain.

The literal source-read scope consists of all 4,844 lines in the eleven text files of the pinned build: the main file, preamble, both figure files, six mathematical section files, and bibliography. In particular the collar proof was read in full, not only its theorem statement. The static companion named in Section 9 is an imported theorem whose complete proof was *not* read for this report. Three actual private originals were read over the explicit intervals in Section 10; they motivate legitimate transfers of operator arguments, not substitute evidence for geometric hypotheses. The sidecar read ledger binds those intervals to file hashes.

1.1 Comparison with current primary literature

The primary arXiv record of Simon [3] describes convergence to a C^0 orbifold with finitely many singularities and continuation in the orbifold sense. This distinction from smooth same-manifold continuation is material. Bamler–Zhang [4] establish heat-kernel, distance, cutoff, and curvature estimates under bounded scalar curvature, including four-dimensional consequences. Buzano–Di Matteo's June 2026 Part II [5] rules out locally Type-I singular behavior under bounded scalar curvature; that statement alone does not exclude every possible concentration rate. Deruelle–Ozuch [6] develop an ALE functional with a boundary correction and a Lojasiewicz inequality. Their result is not automatically the source's unminimized functional or its fixed-tree degenerating-family inequality.

The October 8 literature refresh checked these primary records and their stated scope, not their entire proofs. No query contained confidential private mathematics. No historical priority or novelty claim is made for the contraction principle, generalized-inverse algebra, or Taylor expansion used here. The source-relative contribution is making their precise consequences explicit for this particular collar and its retained Ricci error.

2 The source collar problem, with its actual domains

Let $\Gamma \subset O(4)$ be finite, nontrivial, and free on S^3. Calculations take place on the Euclidean cover and are Γ-equivariant. Write $E = \{r \geq 1\}/\Gamma$ and $C_1 = \{1 \leq r \leq 2\}/\Gamma$. For a shell of radius r, the source's scaled Sobolev norm has the form

$$\|v\|^2_{H^s(r)} = \sum_{j=0}^{s} r^{2j-4} \int_{\widetilde{C}(r)} |\partial^j v|^2 \, \mathrm{d}x, \qquad \|v\|_{H^s_\beta} = \sup_{r=2^j,\ j\geq 0} r^\beta \, \|v\|_{H^s(r)} \, .$$

Fixed shell enlargements give equivalent norms. Filled-space norms also include the ball of radius two. Tensor components here have their unscaled Cartesian size. Set

$$k = 30, \quad 1 < q < 2, \quad X = H^k_q(E; \mathrm{Sym}^2), \quad Y = H^k(C_1; \mathrm{Sym}^2) \times H^{k-2}_{q+2}(E; \mathrm{Sym}^2). \tag{1}$$

The collar map T is restriction to the whole collar, not merely a Dirichlet trace on its inner sphere.

For Euclidean background δ, put $S = D\,\mathrm{Ric}_\delta$. With $(\mathcal{B}v)_j = \partial_i v_{ij} - \frac{1}{2}\partial_j \operatorname{tr} v$,

$$Sv = -\tfrac{1}{2}\Delta v + \tfrac{1}{2}\mathcal{L}_{(\mathcal{B}v)^\#}\delta, \qquad \mathcal{B}Sv = 0, \qquad S\mathcal{L}_V\delta = 0. \tag{2}$$

The overdetermined map is

$$A : X \longrightarrow Y, \qquad Av = (Tv, Sv).$$

A pair $(c, 0)$ need not belong to its range. One must not silently assume surjectivity, remove its diffeomorphism kernel by a scalar coercivity argument, or replace it by a finite square inverse.

2.1 What the inherited extension estimate says

On $1 \le r \le 4R$, the given metric satisfies high-order shell bounds $\|g-\delta\|_{C^{k+2},r} \le \theta_r$, with dyadic $\sum_{r\le R}\theta_r \le \sigma$ small. Neighboring weights are comparable, and $\theta_r \ge \|g-\delta\|_{H^k(C_1)}\, r^{-q}$. With fixed enlargements understood, define

$$m_1 = \sup_{C_1} |\operatorname{Ric} g| + \sum_{r\le R} \sup_{\widetilde{C}(r)} \left(\theta_r |\operatorname{Ric} g| + \sum_{j=1}^{k-2} r^j |\nabla^j \operatorname{Ric} g| \right) + \frac{\theta_R}{R^2}. \tag{3}$$

The source constructs $\bar{g} = \delta + h(c)$, depending analytically only on $c = (g-\delta)|_{C_1}$, such that

$$\|h(c)\|_X \le C\,\|c\|_{H^k(C_1)}, \qquad \|c - Th(c)\|_{H^k(C_1)} + \|\mathrm{Ric}(\delta + h(c))\|_{H^{k-2}_{q+2}} \le C m_1. \tag{4}$$

Interpolation on a fixed collar gives $\widehat{g} = g$ near its inner boundary and $\widehat{g} = \bar{g}$ outside the collar, with the same type of Ricci estimate. These are small-error extensions, not arbitrary Ricci-flat fillings. The map is independent of R, the far charts, and the auxiliary weights. The proof of its error bound is not.

The term θ_R/R^2 in (3) is essential: a finite annulus can carry a homogeneous exterior curvature field even when its Ricci tensor vanishes. Likewise a small constant Ricci mode must occur with the summable factor θ_r; summing its unweighted magnitude over every shell would introduce a spurious neck-length loss.

3 Why the source operator is a collar operator

We first record the specific analytical construction on which the local results depend. The source proves the existence of a bounded map

$$L : Y \longrightarrow X, \qquad ALA = A. \tag{5}$$

This identity, rather than $LAL = L$, is its needed range property. Here are the mechanisms behind it and behind the later absorption estimate.

3.1 Filled-space Newton potentials and cancellation

The scalar Newton operator on $\mathbb{R}^4$, applied componentwise, satisfies

$$\Delta^{-1} : H^{k-2}_{q+2} \longrightarrow H^k_q, \qquad 0 < q < 2.$$

At the faster weight needed for its divergence one uses

$$\int_{\mathbb{R}^4} f \, \mathrm{d}x = 0, \quad f \in H^{k-3}_{q+3}, \quad 1 < q < 2 \quad \Longrightarrow \quad \left\| \Delta^{-1} f \right\|_{H^{k-1}_{q+1}} \le C_q \, \|f\|_{H^{k-3}_{q+3}}. \tag{6}$$

For clarity, the cancellation replaces the kernel $G(x-y)$ by $G(x-y) - G(x)$. In $|y| < |x|/2$ its bound is $C|x|^{-3}|y|$; the truncated first moment of $|f(y)| \lesssim (1+|y|)^{-q-3}$ is $O(|x|^{2-q})$. Thus this part is $O(|x|^{-q-1})$. The remaining regions have the same order by kernel scaling and integrability at infinity when $q > 1$. Differentiation and local elliptic estimates give the stated shell Sobolev orders. This is a *filled-space* argument. An exterior-only zero-integral assertion would leave uncontrolled inner multipoles.

For $d = (v_1, f) \in Y$, extend v_1 into the ball and splice $S(Ev_1)$ to f through the collar to form $f^{\#}$. Let

$$u = \Delta^{-1}(-2f^{\#}).$$

A local potential for $v_1 - u$, with a fixed annular cutoff, supplies the gauge correction $\mathcal{L}_{V_d}\delta$; then $Ld = u + \mathcal{L}_{V_d}\delta$. The local potential estimate uses linearized curvature K:

$$\|v - \mathcal{L}_{\mathcal{P}v}\delta\|_{H^k(C_1)} \le C \|Kv\|_{H^{k-2}(C_1)}, \qquad \|\mathcal{P}v\|_{H^{k+1}} \le C \|v\|_{H^k}.$$

Its source proof uses two Neumann gradient projections of the linearized connection, vanishing of the simply connected annulus's relevant div–curl kernel, and normalization of the remaining affine fields. Averaging over Γ preserves the estimates.

For $d = Av$, use the same filling to obtain a global V with $f^{\#} = SV$. Then $\mathcal{B}f^{\#} = 0$, hence $\mathcal{B}u = 0$ and $Su = f^{\#}$. The tensor $K(V - u)$ is an entire decaying harmonic tensor and vanishes. The collar gauge correction therefore restores its entire collar value. This proves $ALAv = Av$. It does not prove that every $d \in Y$ is admissible Ricci and collar data.

3.2 The nonlinear comparison is not a fixed coordinate shortcut

The source constructs a comparison gauge only to estimate the branch, not to define it. On the finite annulus it has the form

$$F_z(x) = e^{J(x)}(x + U(x)), \qquad J(x)^* = -J(x),$$

with J, U zero on the whole collar, J commuting with the quotient action, $\sup|J| \le C\sigma n(z)$, and positive derivatives of J, and U/r, controlled shellwise by $\theta_r n(z)$. Local rigid rotations are matched by telescoping; derivatives of the partition annihilate common rotations. Nontriviality and freeness of Γ exclude invariant translations. Because e^J is orthogonal, the undifferentiated rotation cancels exactly in the flat pullback. The relevant derivative is

$$(e^J)^{-1}\partial_i e^J = \partial_i J + O(|J|\,|\partial_i J|),$$

so the nonlinear comparison loses $C\sigma\theta_r$, not a remote undifferentiated J^2 term. Replacing this by an arbitrary chart perturbation would lose the source's uniform long-annulus estimate.

For the exact selected branch, write $d_1 = (c - Th, -\mathrm{Ric}(\delta + h))$ and $e = \|d_1\|_Y$. The source's filled comparison and elliptic shell sums eventually give

$$\|(I - AL)d_1\|_Y \le C\{m_1 + (\sigma + R^{-q-2})e\}. \tag{7}$$

The inner extension of the comparison uses the same collar filling as L, so its curvature estimate has no uncancelled inner Ce term. The outer contribution contains both θ_R/R^2 and eR^{-q-2}. Moreover $\mathcal{B}f^{\#}$ has integral zero: its flux at infinity is $O(r^{1-q}) \to 0$, with no inner boundary after filling. Equation (6) is what prevents loss of the faster divergence weight. These details explain why a finite inverse norm cannot stand in for the analytic proof of (7).

4 Exact range algebra and the retained compatibility error

All norms below are the actual Banach norms in (1); projections need not be orthogonal or have norm one.

Proposition 4.1 (Reflexive reduction)**.** *Assume* (5)*. Define*

$$Q = LA : X \to X, \quad P = AL : Y \to Y, \quad B = QL = LAL : Y \to X.$$

Then

$$Q^2 = Q, \quad P^2 = P, \quad AQ = A, \quad AB = P, \quad BA = Q, \quad QB = B, \quad BP = B, \quad BAB = B. \tag{8}$$

The closed space $X_0 = \operatorname{im} Q$ *satisfies* $A : X_0 \to \operatorname{im} P$ *is an isomorphism with inverse* $B|_{\operatorname{im} P}$*. For any* $h \in X_0$*, put*

$$F(h) = (Th, \operatorname{Ric}(\delta + h)) = Ah + \mathcal{N}(h), \qquad \mathcal{N}(h) = (0, \operatorname{Ric}(\delta + h) - Sh).$$

For data $d \in Y$*, set* $E = d - F(h)$ *and* $r = BE$*. Then*

$$PE = Ar, \qquad E = (I - P)E + Ar, \qquad BF(h) = h + B\mathcal{N}(h). \tag{9}$$

In particular a vanishing projected residual does not force $E = 0$*.*

Proof. For example $Q^2 = L(ALA) = LA$, $P^2 = (ALA)L = AL$, $AB = ALAL = AL = P$, and $BA = LALA = LA = Q$. The other identities follow by the same associations; $BAB = QB = B$. A bounded projection has closed range. If $x \in X_0$, $BAx = x$; if $y \in \operatorname{im} P$, $ABy = y$, and $By \in X_0$. Finally $PE = ABE = Ar$, and $BAh = Qh = h$. No relation $LAL = L$ was used. □

For a concrete warning, take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

Here $ALA = A$ but $LAL \neq L$. The nonzero datum $d = (-2, 1)^T$ has $Bd = 0$. This finite example checks the logic of the reduction; it does not model the continuum Ricci range or certify its bounded operator.

For the source datum $d = (c, 0)$, the selected branch solves

$$h + B\mathcal{N}(h) = Bd, \qquad h \in X_0. \tag{10}$$

Its exact residual d_1 has $Bd_1 = 0$, so $ALd_1 = Pd_1 = 0$. Using (7) and then choosing $C(\sigma + R^{-q-2}) < 1$ yields (4). Thus range selection and the geometric compatibility estimate play different roles: neither one can replace the other.

5 The actual nonlinear Ricci term

Indices in this section are raised by δ, and repeated indices are summed from one to four. Let

$$C^a_{ij}(h) = \tfrac{1}{2}(\partial_i h_{aj} + \partial_j h_{ai} - \partial_a h_{ij}).$$

For $g = \delta + h$, the exact connection is $\Gamma^a_{ij} = (I + h)^{-1}_{ab} C^b_{ij}(h)$. Our curvature convention is the one in (2):

$$\operatorname{Ric}_{ij} = \partial_a \Gamma^a_{ij} - \partial_j \Gamma^a_{ia} + \Gamma^a_{ab}\Gamma^b_{ij} - \Gamma^a_{jb}\Gamma^b_{ia}.$$

Consequently its homogeneous quadratic term is

$$\begin{aligned} R_{2,ij}(h) = \; & -\partial_a(h_{ab} C^b_{ij}(h)) + \partial_j(h_{ab} C^b_{ia}(h)) \\ & + C^a_{ab}(h) C^b_{ij}(h) - C^a_{jb}(h) C^b_{ia}(h). \end{aligned} \tag{11}$$

In particular coefficient derivatives and connection products are both present. Put $\mathcal{N}_2(h) = (0, R_2(h))$.

Lemma 5.1 (Weighted nonlinear estimates)**.** *For fixed q, k, Γ as in* (1)*, there is $a_0 > 0$ such that F is analytic on $\|h\|_X < a_0$. The metric is positive definite there. Constants independent of any finite outer radius satisfy, for $\|h\|_X, \|v\|_X \le a \le a_0$,*

$$\|\mathcal{N}(h)\|_Y \le C_2 \|h\|_X^2, \qquad \|D\mathcal{N}(h)\|_{X\to Y} \le C_1 a, \tag{12}$$

$$\|\mathcal{N}(h) - \mathcal{N}_2(h)\|_Y \le C_3 \|h\|_X^3, \qquad \left\|D^j\mathcal{N}(h)\right\| \le C_j' \quad (j = 2, 3), \tag{13}$$

$$\|\mathcal{N}_2(h) - \mathcal{N}_2(v)\|_Y \le C(\|h\|_X + \|v\|_X)\,\|h - v\|_X. \tag{14}$$

Moreover $D^2\mathcal{N}(0)[h, v] = \mathcal{N}_2(h + v) - \mathcal{N}_2(h) - \mathcal{N}_2(v)$.

Proof. Rescale each shell to a fixed annulus. Sobolev embedding and multiplication at $k = 30$ give a uniform C^2 bound and make H^{k-2} an algebra. For sufficiently small a_0, $\|h\|_{L^\infty} < 1/2$. The inverse metric is the convergent Banach-algebra series

$$(I + h)^{-1} = I - h + h^2(I + h)^{-1}.$$

Insert it into the exact connection and curvature formulas. The linear term is Sh, the quadratic term is (11), and every remaining term has at least three factors of h or its derivatives, with at most two derivatives in total and bounded analytic inverse-metric factors. On a shell this gives

$$\|R_2(h)\|_{H^{k-2}(r)} \le Cr^{-2}\|h\|_{H^k(r)}^2, \quad \|\mathrm{Ric}(\delta + h) - Sh - R_2(h)\|_{H^{k-2}(r)} \le Cr^{-2}\|h\|_{H^k(r)}^3.$$

Multiplication by r^{q+2} leaves respectively factors r^{-q} and r^{-2q}, both at most one on E. The same estimates applied to differentiated inverse-metric series prove the derivative and Lipschitz bounds. The collar component of $\mathcal{N}$ is zero. Polarization of a homogeneous quadratic map proves the final identity. This argument takes the supremum over all shells and so does not acquire a factor proportional to their number. □

An independent consistency check is conformal. If $h = t\phi\delta$, write $\delta + h = e^{2\omega}\delta$ with $\omega = \frac{1}{2}\log(1 + t\phi)$. In dimension four,

$$\mathrm{Ric}_{ij} = -2(\omega_{ij} - \omega_i\omega_j) - (\Delta\omega + 2|\nabla\omega|^2)\delta_{ij}.$$

Expansion gives

$$S(\phi\delta)_{ij} = -\phi_{ij} - \tfrac{1}{2}\Delta\phi\,\delta_{ij}, \qquad R_2(\phi\delta)_{ij} = \phi\phi_{ij} + \tfrac{3}{2}\phi_i\phi_j + \tfrac{1}{2}\phi\Delta\phi\,\delta_{ij}. \tag{15}$$

The apparent quadratic $|\nabla\phi|^2\delta_{ij}$ terms cancel. For a genuine exterior test, ϕ can be a smooth radial function supported in a finite shell; a polynomial jet check alone is not a decaying exterior metric.

6 A quadratic collar predictor and its obstruction

The following is a local consequence of the actual L and Ricci nonlinearity. It is not a claim that arbitrary collar data are compatible. Fix a ball of radius $a < a_0$ in X_0, write $\beta = \|B\|$, and decrease a so that

$$\kappa := \beta C_1 a < 1, \qquad \|Bd\|_X \le (1 - \kappa)a. \tag{16}$$

All estimates are for this fixed local neighborhood. The constants depend on the continuum operator and are not asserted computable from a finite sample.

Theorem 6.1 (Explicit second-order branch and full residual)**.** *Under* (16)*, equation* (10) *has a unique solution h_* in the closed a-ball of X_0. For small $\delta_d = \|d\|_Y$, set*

$$z = Bd, \qquad h_2 = z - B\mathcal{N}_2(z).$$

Then

$$\|h_* - h_2\|_X \le C\delta_d^3, \tag{17}$$

$$d - F(h_2) = (I-P)d - (I-P)\mathcal{N}_2(Bd) + O_Y(\delta_d^3). \tag{18}$$

The same expansion (18) *holds for $d - F(h_*)$. In particular $Pd = d$ removes the linear obstruction but not, in general, its quadratic successor.*

Proof. The map $\mathcal{T}_d(h) = Bd - B\mathcal{N}(h)$ maps X_0 to itself because $QB = B$. As $\mathcal{N}(0) = 0$, (12) gives $\|\mathcal{T}_d(h)\| \le \|Bd\| + \kappa a \le a$. Its Lipschitz constant is at most $\kappa < 1$. Completeness proves existence and uniqueness. Also

$$\|h_*\| \le \frac{\beta}{1-\kappa}\delta_d, \qquad \|h_* - z\| \le \beta C_2 \|h_*\|^2 = O(\delta_d^2).$$

Subtract the defining formulas:

$$h_* - h_2 = -B\{\mathcal{N}(h_*) - \mathcal{N}_2(z)\}.$$

The cubic remainder in (13) and the quadratic Lipschitz bound (14) make the braces $O_Y(\delta_d^3)$. For sufficiently small δ_d, h_2 also defines a positive metric within the analytic neighborhood. Using $AB = P$, we have

$$Ah_2 = Pd - P\mathcal{N}_2(z), \qquad \mathcal{N}(h_2) = \mathcal{N}_2(z) + O_Y(\delta_d^3).$$

Subtracting their sum from d proves (18). The local Lipschitz bound for F and (17) transfer it to h_*. □

Corollary 6.2 (Second-order compatible data)**.** *For $d(t) = td^{(1)} + t^2 d^{(2)} + O_Y(t^3)$, the branch has expansion*

$$h_*(t) = tBd^{(1)} + t^2\{Bd^{(2)} - B\mathcal{N}_2(Bd^{(1)})\} + O_X(t^3).$$

Its full residual is $O_Y(t^3)$ if and only if

$$(I-P)d^{(1)} = 0, \qquad (I-P)\{d^{(2)} - \mathcal{N}_2(Bd^{(1)})\} = 0.$$

Proof. Apply Theorem 6.1 and homogeneity of $\mathcal{N}_2$. The coefficients of t and t^2 must vanish separately, as is seen by division first by t, then by t^2, and taking $t \to 0$. □

For collar-only data all $d^{(j)}$ have second component zero. The corollary is then a genuine constraint on collar jets, not permission to choose an arbitrary interior Ricci datum to cancel it. Nor is the obstruction space claimed finite dimensional: Y contains overdetermined differential data, and no such dimension statement follows from (5). The predictor is a rigorous local Taylor refinement, not a measured speedup or a proof that the obstruction vanishes.

7 Inexact reconstruction without changing the source weights

Theorem 7.1 (Residual to exact branch)**.** *Retain* (16)*, and let* $\widetilde{h} \in X_0$ *lie in the same a-ball. Define its* continuum *projected residual by*

$$r = B\{d - F(\widetilde{h})\}, \qquad \varepsilon_h = \frac{\|r\|_X}{1-\kappa}.$$

Then

$$\left\|\widetilde{h} - h_*\right\|_X \le \varepsilon_h, \qquad \left\|[d - F(\widetilde{h})] - [d - F(h_*)]\right\|_Y \le (\|A\| + C_1 a)\varepsilon_h. \tag{19}$$

If $d = (c, 0)$ *comes from a metric satisfying the source collar hypotheses, and* (4) *holds for* h_**, then*

$$\left\|c - T\widetilde{h}\right\|_{H^k(C_1)} + \left\|\mathrm{Ric}(\delta + \widetilde{h})\right\|_{H^{k-2}_{q+2}} \le Cm_1 + (\|A\| + C_1 a)\varepsilon_h. \tag{20}$$

Proof. Equation (9) gives $\widetilde{h} = Bd - B\mathcal{N}(\widetilde{h}) - r$, whereas h_* satisfies the same identity without r. Subtraction yields $\left\|\widetilde{h} - h_*\right\| \le \kappa \left\|\widetilde{h} - h_*\right\| + \|r\|$. This proves the first estimate. The mean value formula for $\mathcal{N}$ on the convex ball proves the second. The triangle inequality and (4) prove (20). In particular no comparison gauge or shell weights were reassigned to $\widetilde{h}$; those hypotheses were used only for the exact branch. □

Corollary 7.2 (An exactly matching inner boundary)**.** *Choose a fixed smooth radial cutoff* ζ*, equal to one near the inner sphere and supported before the outer boundary of* C_1*. For collar perturbation* c*, define*

$$\mathcal{H}(c, h) = h + \zeta(c - Th)$$

using zero extension of the supported collar term. Then $\delta + \mathcal{H}(c, \widetilde{h})$ *equals the given* g *near the inner sphere. For sufficiently small* $c, \widetilde{h}$*,*

$$\left\|\mathcal{H}(c, \widetilde{h}) - \mathcal{H}(c, h_*)\right\|_X \le C_\zeta \varepsilon_h, \qquad \left\|\mathrm{Ric}(\delta + \mathcal{H}(c, \widetilde{h}))\right\|_{H^{k-2}_{q+2}} \le Cm_1 + C\varepsilon_h.$$

Proof. Multiplication by ζ and restriction T are bounded between the displayed fixed-collar spaces. Thus the difference is $(I - \zeta T)(\widetilde{h} - h_*)$, with norm at most $C_\zeta \varepsilon_h$. Smallness preserves positive definiteness of both interpolated metrics. Apply the local Ricci Lipschitz estimate on that slightly enlarged small ball, then the inherited bound for the interpolated exact branch. □

If a computed candidate is not in X_0, first take $h_0 = Q\widetilde{h}$. Provided h_0 is in the admissible ball, the valid bound is

$$\left\|\widetilde{h} - h_*\right\|_X \le \left\|(I - Q)\widetilde{h}\right\|_X + \frac{\left\|B[d - F(Q\widetilde{h})]\right\|_X}{1-\kappa}. \tag{21}$$

The first term cannot be discarded. If the data or residual are evaluated approximately, verified bounds on their errors in Y or X_0, respectively, must be added before division by $1-\kappa$. A coefficient residual at finitely many modes or points is not the continuum residual in this theorem.

7.1 The direct compatibility inequality, with its missing hypotheses exposed

There is also an elementary conditional version of the source absorption. For any $E = d - F(\widetilde{h})$, Proposition 4.1 gives $PE = Ar$. If one has independently established

$$\|(I - P)E\|_Y \le Cm + \alpha \|E\|_Y, \qquad 0 \le \alpha < 1,$$

then

$$\|E\|_Y \le \frac{Cm + \|A\|\,\|r\|_X}{1-\alpha}. \tag{22}$$

Indeed use (9), apply the triangle inequality, and absorb. For the source, $\alpha = C(\sigma + R^{-q-2})$, but applying this comparison directly to an arbitrary inexact metric is not automatic. Its extra shell amplitude is $O(\varepsilon_h r^{-q})$. If the original comparison is to be rerun, its weight must dominate that term, for example $\theta'_r = \theta_r + C\varepsilon_h r^{-q}$; the sum, smallness threshold, weighted Ricci term, and outer-boundary term must all be recalculated with θ'. Alternatively one must assume the candidate is already small relative to every original collar weight. Equation (20) avoids this unproved transfer by comparing to the exact selected branch.

8 Parameter derivatives and the exterior variation budget

Proposition 8.1 (Full derivative of the selected metric)**.** *Let $d(t)$ be C^2, stay in a smaller data neighborhood satisfying* (16)*, and let $h(t) = h_*(d(t))$. Put*

$$J_h = I_{X_0} + BD\mathcal{N}(h)|_{X_0}.$$

Then

$$\dot h = J_h^{-1} B\dot d, \tag{23}$$

$$\ddot h = J_h^{-1} B\{\ddot d - D^2\mathcal{N}(h)[\dot h, \dot h]\}. \tag{24}$$

If $\|D^2\mathcal{N}\| \le M_2$, then

$$\left\|\dot h\right\| \le \frac{\beta}{1-\kappa}\left\|\dot d\right\|, \qquad \left\|\ddot h\right\| \le \frac{\beta}{1-\kappa}\left\|\ddot d\right\| + \frac{\beta^3 M_2}{(1-\kappa)^3}\left\|\dot d\right\|^2.$$

For $d \in W^{1,1}(I;Y)$, represented by the integral of a Bochner-integrable derivative, (23) *holds almost everywhere, with the corresponding integrable norm estimate.*

Proof. The Neumann series for J_h^{-1} converges since $\|BD\mathcal{N}(h)\| \le \kappa < 1$, and has norm at most $(1-\kappa)^{-1}$. Differentiating (10) once and twice gives the formulas, with the displayed second derivative term retained. Analytic implicit inversion, or the contraction construction followed by difference quotients, makes these differentiations legitimate. For the last assertion, the solution map is continuously differentiable with bounded derivative on the smaller neighborhood. The Sobolev chain rule for a path with the supplied Bochner derivative gives the formula almost everywhere and its integral representation. Arbitrary Banach-space absolute continuity alone would not justify an almost everywhere Bochner derivative; no Radon–Nikodym property is assumed for the weighted shell-supremum space. □

Only collar data are differentiated here. Time derivatives of far comparison charts, auxiliary weights, or R are not part of (23). The fixed-material-collar and patch-switching construction in the global source is what makes this distinction useful. For an approximate path, a small state residual alone says nothing about its velocity: small-amplitude arbitrarily fast oscillations are a counterexample. A velocity estimate needs its own residual or derivative control.

Proposition 8.2 (Physical-scale tail error)**.** *Fix $q > 1$ and a physical collar radius $b > 0$ for a time patch. Suppose two positive exterior metrics, one exact-selected and one inexact, have equivalent volume density bounded by $Cb^4 r^3\,\mathrm{d}r\,\mathrm{d}\omega$. Suppose their inexact Ricci and velocity satisfy*

$$|\operatorname{Ric}|_g \le Cb^{-2}(m+\varepsilon)r^{-q-2}, \qquad |\dot g|_g \le C(v+\varepsilon_1)r^{-q} \quad (r \ge 1),$$

where all four coefficients are nonnegative. The absolute exterior variation pairing obeys

$$\left| \int_E \left\langle \tfrac{1}{2} R g - \operatorname{Ric}, \dot{g} \right\rangle_g \, \mathrm{d}V_g \right| \le \frac{C b^2}{2q-2} (m+\varepsilon)(v+\varepsilon_1). \tag{25}$$

Relative to its exact allowance $Cb^2 mv/(2q-2)$, the additional allowance consists of

$$\frac{C b^2}{2q-2} (m\varepsilon_1 + v\varepsilon + \varepsilon\varepsilon_1).$$

For time-dependent coefficients, the same bound integrates in time, and each product is controlled by the product of its L^2 norms.

Proof. In dimension four $|R| \le 2|\operatorname{Ric}|$, so the integrand is bounded by $C|\operatorname{Ric}||\dot{g}|$. Multiply the three assumed bounds. The radial integral is $\int_1^\infty r^{1-2q}\,\mathrm{d}r = (2q-2)^{-1}$. Expansion of the two factors gives the allowance; Cauchy–Schwarz proves its time-integrated form. □

For the original source v is controlled by the collar quantity $B_1(t)$. Theorem 7.1 and Corollary 7.2 give a state error $\varepsilon = C\varepsilon_h$ at unit collar scale. Supplying a separately proved ε_1 then makes (25) an explicit admissible solver-error budget in the source's energy argument. It is not a global certificate until its time integral is small compared with that argument's a^4-scale terms. The divergence as $q \downarrow 1$ explains a real endpoint obstruction.

9 What the collar estimates do not prove about the flow

The rest of the source requires several logically independent geometric and parabolic steps. We record their mechanisms and quantifiers because they delimit the local results above.

9.1 Concentration, genuine quotient tips, and the root scale

The source defines a spatial curvature radius $\ell(x,t)$ using curvature control on whole balls, capped at a fixed small radius. It is spatially Lipschitz and time-continuous. Its bounded-scalar analysis supplies local volume comparability, a uniform slice L^2-curvature bound, derivative estimates

$$|\nabla^j \operatorname{Rm}| \le C_j \ell^{-2-j}, \qquad |\nabla^j \operatorname{Ric}| \le C_j \ell^{-1-j},$$

and a uniformly finite covering of the set $\{\ell < r\}$ by Cr-balls. In particular its volume is $O(r^4)$.

These are not consequences of the collar inverse. The source obtains heat estimates from entropy and scalar control, constructs spacetime cutoffs, and first establishes curvature-radius time control on a nonempty interval before bootstrapping it. The improved Ricci scale comes from integrating the scalar evolution over a curvature-scale cylinder, then applying the homogeneous Ricci tensor equation. Four-dimensional Gauss–Bonnet and a scalar-denominator Ricci energy estimate enter the slice curvature bound.

Its blowup limits are complete noncollapsed Ricci-flat orbifolds with regular smooth convergence. Energy entry and volume arguments give a finite bubble tree. The proof that a genuine tip has a nontrivial group free on S^3 matters twice: it excludes invariant translations in the collar gauge, and gives strict cone-link diameter below π for geodesic estimates. A limiting Ricci-flat space need not be flat. Likewise removal of a puncture uses harmonic-coordinate elliptic regularity with improving integrability, not merely small scalar curvature at that puncture.

On a long neck the source assumes two-ended Ricci decay, selects $0 < \mu < \min\{1, \sigma_0\}$, and proves matching full-curvature and metric decay. The possible scale-critical linear curvature mode

is removed by the uncontracted Bianchi identity. The metric estimate then requires local Saint–Venant/Korn potentials and alignment of rigid motions. A scalar elliptic estimate alone misses those compatibility and gauge steps.

The root scale $a(t)$ is a continuous cutoff maximum of the curvature radius times a curvature detector. It measures the outermost nonflat concentration, not the smallest bubble. Short-motion control operates on a physical interval $o(h)$, not only on an interval $o(h^2)$: the improved Ricci estimate is essential here. Buffered intervals around a fixed-factor change of a are selected so that their rescaled time lengths diverge in curvature-scale units. After rescaling by their physical duration, one has $\varepsilon_i \to 0$, $a_i \to 0$, and exterior weights

$$\Theta_i(r) = (a_i/r)^\mu + \delta_i r^\mu, \qquad \delta_i = (R_i^{\mathrm{out}})^{-\mu}, \qquad \varepsilon_i R_i^{\mathrm{out}} \to 0.$$

The exterior geometry is established before a large cutoff N is chosen. Otherwise a hidden N-dependent constant can invalidate the final absorption.

9.2 Cutoff Ricci, angular modes, and positive derivatives

Let $G = \eta(t)\,\mathrm{Ric}$, with η compactly supported in a buffered time interval and positive on its central interval. For lower-index Ricci the evolution includes both

$$(\partial_t - \Delta)\,\mathrm{Ric} = 2\,\mathrm{Rm} * \mathrm{Ric} - 2\,\mathrm{Ric}^2, \qquad (\partial_t - \Delta)G = \text{homogeneous terms} + \eta'\,\mathrm{Ric}\,.$$

The source estimates the last term directly; division by η would be invalid at its support boundary. Its shell rescaling $G_r = r^{-2}\Phi_r^* G$ records the original tensor magnitude, and is not the Ricci tensor of the rescaled metric.

Subtracting the componentwise spherical mean in Cartesian coordinates commutes with the flat heat operator. The first nonconstant spherical eigenvalue on S^3 is three, giving the radial comparison operator

$$\mathcal{L}z = -z'' - \frac{3}{r}z' + \frac{3}{r^2}z.$$

Its indicial roots are $1, -3$, since substituting r^λ gives $-\lambda^2 - 2\lambda + 3 = 0$. The trace-reversed contracted Bianchi identity recovers derivatives of the mean from the oscillation and metric error. In particular a constant matrix coefficient in $r^{-2}A$ is not an undetected mean mode: the divergence of its trace reverse is

$$-2r^{-3}(A - \tfrac{1}{2}(\operatorname{tr} A)I)n,$$

which vanishes for every $n \in S^3$ only if $A = 0$ in dimension four. An actual constant tensor can remain; it is the positive spatial derivatives, rather than this constant magnitude, that are summable over the entire exterior.

Writing K for the normalized cutoff-Ricci energy in a root-scale ball, the source obtains schematic estimates of the precise forms

$$V(r) \le C\{(a/r)^{2+\alpha}K + \widetilde{\delta}[(a/r)^\alpha + r^\alpha]\},$$
$$A_0(r) \le C\{(a/r)^{2+\alpha}K + \widetilde{\delta}\}, \qquad 0 < \alpha < \min\{\mu, 1\}.$$

Here V is a time-L^2 norm of positive scaled derivatives, whereas A_0 controls the zero-order magnitude. The proof chooses a large shell ratio first, a large parabolic past margin next, geometric cutoffs next, and finally the sequence index. The leading recurrence constants must not depend on the shell ratio or past margin.

For deep tree regions the source uses an L^p-time maximal envelope, $1 < p < 2$, whose L^2 norm is bounded by $C(K + o(1))$. When a time average over a large scale B^2 is transferred to a local scale s^2, the factor $(B/s)^{2/p}$ must be retained. The chosen neck parameters make the resulting exponents 2λ, $(4 - 2/p)\lambda$, $(2 - 2/p)\lambda$, and $\delta - (2/p)\lambda$ positive. Omitting this factor would make the asserted decay easier but would not prove the source estimate. The uncut Ricci smallness used by the static theorem is a separate qualitative estimate, not obtained by dividing the cutoff bound by η.

9.3 The static functional and the nonuniform little-o input

On an ALE exterior in four dimensions the source uses

$$\mathcal{E}(g) = \lim_{D\to\infty} \left[\int_{\{r<D\}} R_g \, \mathrm{d}V_g - \int_{\{r=D\}} (\partial_j g_{ij} - \partial_i g_{jj}) n_i \, \mathrm{d}S_\delta \right]. \tag{26}$$

The two terms need not converge separately. The linear divergence in the scalar density cancels against the displayed boundary term. For $g - \delta = O(r^{-q})$, the remaining radial bound is r^{1-2q}, integrable precisely in the stated range $q > 1$. Its variation and physical scaling are

$$D\mathcal{E}_g(\dot g) = \int \langle \tfrac{1}{2} R g - \mathrm{Ric}, \dot g \rangle_g \, \mathrm{d}V_g, \qquad \mathcal{E}(s^2 g) = s^2 \mathcal{E}(g)$$

with the corresponding dilation of asymptotic coordinates. Coordinate invariance requires a decaying displacement and control along the whole isotopy, not just a formal equality at its endpoints.

The imported companion [2] concerns a *fixed finite* rooted Ricci-flat tree, fixed matching charts and quotient groups, joined scales $a_c = a_p e^{-2L}$, joined positive decay rates, and a root exterior rate $q > 1$. Its hypotheses include uniform metric derivatives through order twenty and Ricci derivatives through order ten in the specified scaled weights. If $M \to 0$ controls those Ricci errors, its conclusion is

$$\mathcal{E} = o(M).$$

This is stronger than the $O(M)$ estimate obtained directly from variation along a scaling direction. It is not a uniform power law over all trees, and it is not established here. Analytic path selection in the companion does not mean the spatial metrics must be analytic, nor can one replace it by assuming unobstructed Einstein deformations.

9.4 The order of limits and the final contradiction

The source attaches the collar exterior at $b = Na$, first with N fixed and then with the sequence index tending to infinity. The material collar must bound the component containing the true root core; small coordinate metric error alone does not identify that component. The time patches have widths measured in $\int (1 + B_1(t))\, \mathrm{d}t$, and their partition obeys $\sum_j |\chi_j'| \le C(1 + B_1)$ almost everywhere. Their number need not be uniformly bounded.

For the common collar error m and collar velocity B_1, the source proves

$$b^{-2} \, \|\eta m\|_{L^2} + \|\eta B_1\|_{L^2} \le C\{(a/b)^{2+\alpha} K + \rho_i\}, \qquad \rho_i \to 0.$$

The zero-order exterior term is multiplied by its summable metric weight, while positive derivatives are summed directly. The term $\theta_R/(Rb)^2$ is kept until its vanishing is proved. Inside the actual flow region,

$$D\mathcal{E}_g(-2\,\mathrm{Ric}) = \int (2|\,\mathrm{Ric}\,|^2 - R^2)\, \mathrm{d}V_g.$$

The root core supplies the positive a^4K^2 term. The collar and tail cost is bounded by $Cb^2 \int mB_1$, hence by $Ca^4N^{-2\alpha}K^2$ plus sequence errors.

The imported static inequality is applied after extracting a fixed tree along any offending sequence of patch times. The root remains nonflat and its new exterior has decay $q > 1$. This sequential argument yields the uniform little-o ratio needed on active patches without asserting a uniform exponent over all trees. Integration by parts in $\eta^2\chi_j\mathcal{E}_j$ includes both $2\eta\eta'\chi_j$ and $\eta^2\chi_j'$. One cannot identify the different $\mathcal{E}_j$ across patch switches.

Choosing N large enough first absorbs $CN^{-2\alpha}K^2$; only then does the index tend to infinity. The resulting inequality has the form $K^2 \le \gamma_i(K+1)^2$, $\gamma_i \to 0$. It implies

$$K \le \frac{\sqrt{\gamma_i}}{1-\sqrt{\gamma_i}} \longrightarrow 0,$$

with no prior boundedness assumption on K. The source turns the resulting small L^2-time Ricci envelope into distance stability by excluding bad labels of small volume. Its geodesic estimates use the genuine quotient-tip geometry. Good label nets then identify the two pointed root limits isometrically. Persistence of the stopped curvature radius on regular points, together with genuine concentration at tips, contradicts the prescribed fixed-factor change of the root scale.

Only after this contradiction rules out unbounded full curvature does ordinary smooth Ricci-flow continuation on the original manifold apply. The local predictor, an inexact collar residual bound, or small scalar curvature by itself proves none of these last global implications.

10 Private proof lineage and legitimate transfers

The private corpus was read locally. The following are specific proof transfers, not title-based claims of applicability or inherited contribution scores.

10.1 PDE-first formulation: domains and boundary lifts

Report 8 [7] was read at lines 53–178, 196–224, and 270–387 of its original TeX. Its weak Dirichlet proof uses a bounded connected Lipschitz domain, a coercive principal coefficient, and the appropriate H_0^1 realization. It does not identify every weak operator domain with H^2. Its Neumann proof retains the compatibility condition between the volume forcing and outward conormal flux, with one constant mode per connected component. The boundary-lifting identity is the full

$$u = \ell + G_D(g - \mathcal{L}\ell),$$

not an unrestricted free-space potential. The heat lifting contains both $-\partial_t\ell$ and $-\mathcal{L}\ell$. The semilinear integral reformulation becomes a contraction only after an invariant ball and a norm bound are supplied. Its quasilinear derivative includes all coefficient derivatives.

These proofs justify insisting on the original domains, the full nonlinear derivative, and the entire collar correction in this report. They do *not* prove coercivity of S, eliminate its gauge kernel, or provide the source's weighted L. The private cubic H^1 example has a dimension restriction at most three; it cannot simply be transferred to a four-dimensional collar. Here the high Sobolev order in (1) is what licenses multiplication.

10.2 Local–global graph operators: exact residuals

The original core pattern 5 [8] was read at lines 88–133, 189–308, and 364–464. Its physical-metric adjoint and gauge constraints show why a numerical Euclidean pseudoinverse is not automatically

the physical inverse. Its Neumann truncation for $A = D(I - M)$ retains the ordered residual $DM^{K+1}D^{-1}b$; spectral radius alone does not give the required nonnormal operator-norm estimate. Its block elimination proof gives

$$r_{\text{boundary}} = r_{\text{Schur}} + A_{BI}A_{II}^{-1}r_{\text{interior}},$$

including the affine forcing reconstruction. Its finite-rank update proof similarly retains both the original solve error and the update solve error. These are complete algebraic proofs in their stated finite or bounded-operator settings.

The legitimate transfer is to expose the exact projector residual (9), not to identify the nonlinear Ricci collar with a graph Schur complement. In particular an interior solve error cannot be discarded, and finite reconstruction cannot remove the continuum compatibility component $(I - P)E$. The source's filled-space zero-mass cancellation is additional analytic information.

10.3 Stability: right spaces and both derivative terms

The original stability module [9] was read over the bounded proof interval 1–221 of its 295 lines. It proves inverse perturbation estimates in declared norms, includes both terms in a parameter-dependent reconstructed state derivative, and gives a contraction-ball theorem with an invariant radius. Its conjugacy arguments require bounded isomorphisms; a finite frame requires range consistency and does not control discarded modes by itself.

The relevant distinction is that a left-inverse residual $BF = 0$ does not imply $F = 0$ for a rectangular or overdetermined map. Proposition 4.1 makes this explicit for the source's actual $B = LAL$, and Theorem 7.1 bounds distance to its selected branch. The preserved residual in Y is then bounded using the independent geometric estimate, not declared zero. The actual derivative $D\,\mathrm{Ric}$, its second derivative, and the inner-boundary interpolation appear in Sections 5–8. No bounded coordinate change is claimed to create coercivity or remove compactness obstructions.

11 Benefits, costs, and further work

11.1 Benefits specific to this source

The exact reduction $B = LAL$ separates a solver residual from a compatibility residual without assuming an unavailable generalized inverse identity. The explicit R_2 formula gives an implementable second-order local predictor in the same gauge-selected space as the source. Its $O(\|d\|^3)$ error and its quadratic obstruction are proved with the actual inverse-metric derivatives. This is more informative than a statement that some nonlinear map is smooth.

The residual theorem permits approximate construction while preserving the source's full Ricci estimate and an exactly matched inner boundary. Comparing to the exact branch avoids silently assuming that a numerical error respects all source shell weights. Parameter derivatives specify the velocity needed in the energy pairing. The tail allowance then states exactly how state and velocity errors interact at physical scale b, including the non-negligible cross term.

11.2 Costs and limitations

The constants depend on the continuum L, weighted multiplication, the fixed collar cutoff, and the small neighborhood. No computable uniform values over all quotient groups or degenerating trees are provided. The projection is generally oblique, so its norm cannot be replaced by one.

The predictor requires applications of B and actual quadratic derivatives; the proof supplies no operation count, measured speedup, or oracle for these continuum operations.

The weight window $1 < q < 2$ has two different boundaries: the upper endpoint encounters the Newton potential's leading multipole, while the lower endpoint affects zero-mass improvement and makes the variation integral logarithmically divergent. High regularity $k = 30$ leaves room for the source's static derivative requirements; the report does not optimize it. Smallness and positive definiteness are explicit local hypotheses, not consequences of a small sampled residual. Spatially or temporally unobserved modes remain possible.

Finally the collar estimate is only one step in the extension proof. Root selection, genuine tip geometry, exterior constants independent of N, the static little-o inequality, and uniformity along changing time patches are global costs that remain. None is discharged by the finite checks accompanying this report.

11.3 Concrete extensions and their obstacles

A first practical continuation is a validated realization of B on an exterior discretization. It would require an analytic truncation estimate for all omitted shells and angular modes, a bound on $ALA - A$ in the stated spaces, and an explicit filled-space zero-mass correction. A finite matrix condition number alone supplies none of these. Approximate range identities would introduce additional terms in both $Q^2 - Q$ and the equation on X_0; one would need a stable nearby continuum splitting before using the present theorem verbatim.

A second direction is higher-order collar prediction. Analyticity allows recursive Taylor coefficients, but each order has its own $(I - P)$-compatibility condition. Vanishing at second order does not establish unobstructedness. A proof of all-order compatibility for a special geometric family would be a substantive further result.

A third direction is to approach $q = 1$ or $q = 2$ by introducing explicit asymptotic coefficients and renormalized moment conditions. At $q = 1$ the tail pairing and relevant moments are borderline; at $q = 2$ a Newton monopole may survive. One would have to alter the domain and boundary functional, not just substitute the endpoint into the present inequalities.

A fourth direction is certified time-dependent approximation on material collars. State residuals, differentiated residuals, and patch-switching errors should be bounded together so that the integral in Proposition 8.2 is $o(a^4)$ at the needed stage. Uniform estimates must be established before increasing N. This could make the local construction quantitatively robust, but would still leave the static tree theorem and concentration argument as separate mathematical obligations.

12 Contribution, evidence, and boundaries of verification

The catalogue's applicability level is L4. The realized contribution assessment is C2, not L4 relabeled as C4. The new scoped results are Proposition 4.1, Lemma 5.1, Theorem 6.1, Corollary 6.2, Theorem 7.1, Corollary 7.2, and Propositions 8.1–8.2. Their proofs use classical algebra and analysis, with explicit source-specific operators, derivatives, boundary reconstruction, and error budgets. No independent global extension proof or historical novelty claim is attached to them.

The accompanying checks test finite projector identities including $ALA = A$ without $LAL = L$, the conformal quadratic Ricci formula, obstruction and predictor expansions in finite fixtures, and the numerical exponents and scaling identities actually used here. They are sanity checks with a declared scope, not certificates of a continuum inverse, a whole PDE proof, or a global Ricci-flow theorem. Their outcome and binding source hash are recorded in the sidecar check record rather than conflated with the mathematical hypotheses.

The read-scope sidecar records literal intervals and pre-read versus final hashes. Matching endpoint hashes do not prove that a file could not have changed and changed back during a read. The source-identity record separately compares current local text bytes, as Git blobs, to the authenticated October 8 cached base and target trees. This is a dated identity check, not a fresh live-branch retrieval and not a proof check. The full source read, the bounded private proof reads, and the primary-record literature refresh are separate evidence categories. Neither source length nor the number of successful finite checks measures the scope of formal verification.

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